UNION PUBLIC SERVICE COMMISSION

Civil Services (Preliminary Exam) - 2022

CSAT : PAPER-II

(Analysis with Explanation)

Time : 2 hours **Maximum Number : 200**

Directions for the following 3 (three) items

Read the following two passages and answer the items that follow the passages. Your answers tothese items should be based on the passages only.

Passage -1

The main threat to maintaining progress in human development comes from the increasingly evident unsustainability of production and consumption patterns. Current production models rely heavily on fossil fuels. We now know that this is unsustainable because the resources are finite. The close link between economic growth and greenhouse gas emissions needs to served for human development to become truly sustainable. Some development countries have begun to alleviate the worst effects by expanding recycling and inverting in public transport and infrastructure. But most developing countries are hampered by the high costs and low availability of clean energy sources. Developed countries need to support developing countries' transition to sustainable human development.

1. Unsustainability in production pattern is due to which of the following?

1. Heavy dependence on fossil fuels

2. Limited availability of resources

3. Expansion of recycling

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 only

(c) 1 and 3 only (d) 1, 2 and 3

Ans. (a) : As given in the passage, the present production model is heavily dependent on fossil fuels. Along with this, the matter of limited resources has also been said in the passage. Both of these are said to be unsustainable in the nature of production. Hence statement 1 and 2 are true. At the same time, the spread of recycling is not related to the unsustainable nature but is related to the sustainable nature. Hence statement 3 is false.

2. Consider the following statements :

Developed countries can support developing countries transition sustainable human development by

1. Making clean energy sources available low cost

2. Providing loans for improving their public transport at nominal interest rates

3. Encouraging them to change their production and consumption patterns

Which of the statement given above is/are correct

(a) 1 only (b) 1 and 2 only

(c) 2 and 3 only (d) 1, 2 and 3

Ans. (a) : In the passage, the author has clearly mentioned that developing countries have to face high prices and low availability of refined energy. Therefore, developed countries can lead developing countries towards sustainable human development by producing refined energy sources at low cost. Hence statement 1 is correct.

Passage-2

Unless the forces and tendencies which are responsible for destroying the country's environment are checked in the near future and afforestation of denuded areas is taken up on a massive scale, the harshness of the climatic conditions and soil erosion by wing by wind and water will increase to such an extent that agriculture, which is the mainstay of our people, will gradually become impossible. The desert countries of the world and our own desert areas in Rajasthan are a grim reminder of the consequences of large-scale deforestation. Pockets of deserts-like landscape are now appearing in other parts of the country including the Sutlaj-Ganga Plains and the Deccan Plateau. Where only a few decades back there used to be lush green forests with perennial streams and springs, there is only brown earth, bare of vegetation without any water in the streams and springs except in the rainy season.

3. According to the passage given above deforestation and denudation will ultimately lead to which of the following:

1. Depletion of soil resource

2. Shortage of land for the common man

3. Lack of water for cultivation

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

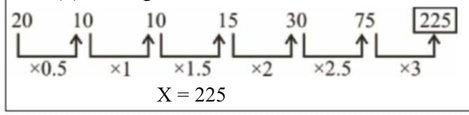
Ans. (c) : According to the passage given above, deforestation and denudation will ultimately lead to–
– Depletion of soil resource
– Lack of water for cultivation.

4. What is the value of X in the sequence 20, 10, 10, 15, 30, 75, X ?

(a) 105 (b) 120

(c) 150 (d) 225

Ans. (d) : The given series is as follows–



X = 225

5. An identity Card has the number ABCDEFG not necessarily in that order, where each letter represents a distinct digit (1,2,4,5,7,8,9 only). The number is divisible by 9. After deleting the first digit form the right, the resulting number is divisible by 6. After deleting two digits from the right of original number, the resulting number is divisible by 5. After deleting three digits from the right of original number, the resulting number is divisible by 4. After deleting four digits from the right of original number, the resulting number is divisible by 3. After deleting five digits from the right of original number, the resulting number is divisible by 2. Which of the following is a possible value for the sum of the middle three digits of the numbers?

(a) 8 (b) 9

(c) 11 (d) 12

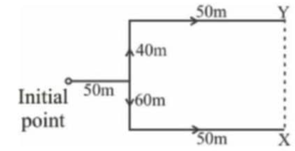
Ans. (a) : Arranged, according to the question
Number = 7412589
Required value = 1 + 2 + 5 = 8

6. Two friends X and Y start running and they run together for 50 m in the same direction and reach a point. X turns right and runs 60 m, while Y turns left and runs 40 m. Then X turns left and runs 50 m and stops, while Y turns right and runs 50 m and then stops. How far are the two friends from each other now?

(a) 100 m (b) 90 m

(c) 60 m (d) 50 m

Ans. (a) : According to the question,



Distance between X and Y = 100 meter

7. Which date of June, 2099 among the following is Sunday?

(a) 4 (b) 5

(c) 6 (d) 7

Ans. (d) : Number of odd days in 2098
= 2000 + (24 leap year) + 74 (Normal year)
= 0 (odd days) + 48 (odd days) + 74 (odd days)
= 122 (odd days)
= 3 (odd days) -------(i)
In June 2099
= January + February + March + April + May + June
= 3 + 0 + 3 + 2 + 3 + June
= 4 (odd days) + June-------(ii)
Since, June 2099 will be a Sunday
= 7 + 7 = 14 = 0 (odd days)
So, 7th June 2099 will be Sunday.

8. A bill for ₹ 1840 is paid is the denominations of ₹ 50, ₹ 20 and ₹ 10 notes, 50 notes in all are used. Consider the following statements :

1. 25 notes of ₹ 50 are used and the remaining are in the denominations of ₹ 20 and ₹ 10

2. 35 notes of ₹ 20 are used and the remaining are in the denominations of ₹ 50 and ₹ 10

3. 20 notes of ₹ 10 are used and the remaining are in the denominations of ₹ 50 and ₹ 20

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d) : According to the Ist statement received formula from ₹50 = 50 × 25 = ₹1250
Remaining money = 1840 – 1250 = ₹590
If remaining 25 note of ₹20, then received money = 25 × 20 = 500
So, ₹1840 is not possible to received the bill.
According to the 2nd statement–
Received money from ₹20 = 20 × 35 = ₹700
Remaining money = 1840 – 700 = ₹1140
If remaining 15 note of ₹50 then received money
= 15 × 50 = ₹750
So, also this is not possible to received the bill.
And according to the 3rd statement–